PHYSICS

Standard Level

Set B | Paper 1A and 1B

Challenge mock examination

Time allowed: 90 minutes Maximum mark: 45

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A calculator and a clean current Physics data booklet are permitted throughout.

• Paper 1A and Paper 1B share one uninterrupted examination time. Paper 1A has 25 multiple-choice questions; Paper 1B has two compulsory experimental/data questions worth 20 marks altogether.

• Answer all questions. Show physical reasoning, working and units in written responses.

• Use the constants and assumptions given in a question. Otherwise use the data booklet and retain unrounded intermediate values.

• Give numerical answers to an appropriate number of significant figures. Diagrams are schematic unless explicitly stated to be to scale.

• The number in square brackets gives the maximum mark for a written part.

Thriving Scholars challenge practice | Current Physics syllabus | 2026

Paper 1A | Question paper

Paper 1A - Multiple choice

Select one answer, A, B, C or D, for each question. Use the answer grid or record your choices on the portal.

1	2	3	4	5	6	7	8	9	10

11	12	13	14	15	16	17	18	19	20

21	22	23	24	25

1A Question 1 [1 mark]

A package is released from a descending platform with initial downward speed $4.0\,\mathrm{m\,s^{-1}}$. Air resistance is negligible; use $g = 10\,\mathrm{m\,s^{-2}}$. What is the ratio of the distance travelled during the third second to that during the first second?

A. $34/14$

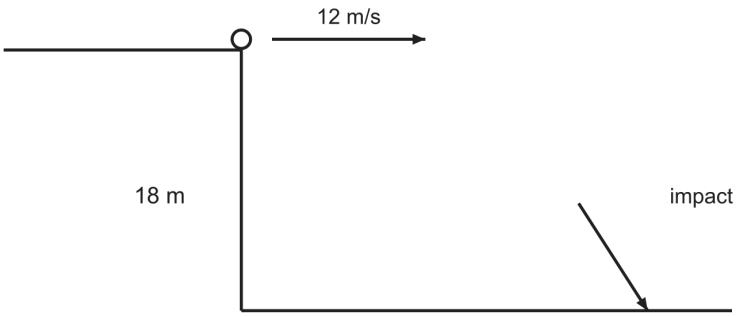
B. 5

C. $29/9$

D. 3

1A Question 2 [1 mark]

A ball leaves a horizontal ledge at $12\,\mathrm{m\,s^{-1}}$. The ground is $18\,\mathrm{m}$ below the ledge. Neglect drag and use $g = 10\,\mathrm{m\,s^{-2}}$. At what angle below the horizontal does it hit?



A. 32°

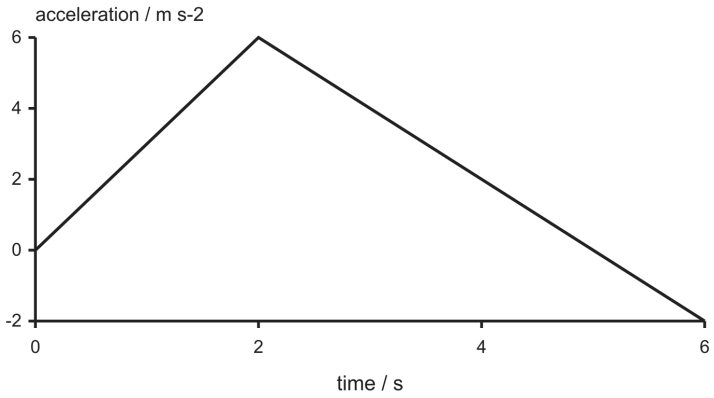
B. 72°

C. 45°

D. 58°

1A Question 3 [1 mark]

The graph gives the acceleration of a $3.0\,\mathrm{kg}$ trolley. The segments are straight. What is the average resultant force over the first $6.0\,\mathrm{s}$?



A. $18\,\mathrm{N}$

B. $7.0\,\mathrm{N}$

C. $9.0\,\mathrm{N}$

D. $12\,\mathrm{N}$

1A Question 4 [1 mark]

Carts of masses m and $3m$ approach one another on a frictionless line with equal speeds v. They stick together. What fraction of the original total kinetic energy remains as kinetic energy?

A. $1/4$

B. 0

C. $3/4$

D. $1/2$

1A Question 5 [1 mark]

A pump raises water at $12\,\mathrm{kg\,s^{-1}}$ through $5.0\,\mathrm{m}$. Water enters nearly at rest and leaves at $2.0\,\mathrm{m\,s^{-1}}$. The useful mechanical-energy gain is 60% of electrical input. Use $g = 10\,\mathrm{m\,s^{-2}}$. What input power is required?

A. $0.374\,\mathrm{kW}$

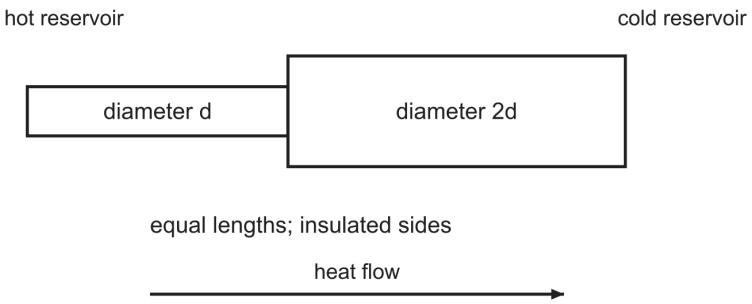
B. $0.624\,\mathrm{kW}$

C. $1.00\,\mathrm{kW}$

D. $1.04\,\mathrm{kW}$

1A Question 6 [1 mark]

Two rods of the same material and length are joined end to end between fixed-temperature reservoirs. Their diameters are d and $2d$. The thinner rod is replaced by another rod of diameter $2d$, keeping both lengths unchanged. Neglect losses and contact resistance. By what factor does the steady heat-transfer rate increase?



A. 4

B. 1.6

C. 2.5

D. 2

1A Question 7 [1 mark]

At the same liquid temperature, a $60\,\mathrm{W}$ heater produces a warming rate of $6.0\,\mathrm{K\,min^{-1}}$ and a $100\,\mathrm{W}$ heater produces $14\,\mathrm{K\,min^{-1}}$. The liquid mass is $0.20\,\mathrm{kg}$; container heat capacity is negligible. Heat loss depends only on temperature. What is the liquid's specific heat capacity?

A. $1500\,\mathrm{J\,kg^{-1}\,K^{-1}}$

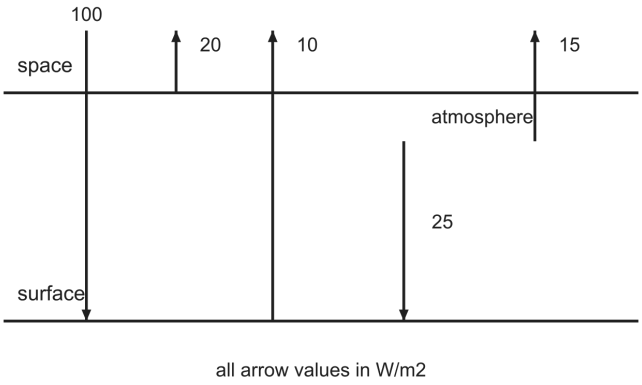
B. $2140\,\mathrm{J\,kg^{-1}\,K^{-1}}$

C. $300\,\mathrm{J\,kg^{-1}\,K^{-1}}$

D. $500\,\mathrm{J\,kg^{-1}\,K^{-1}}$

1A Question 8 [1 mark]

The model is in steady state. Incoming solar intensity is $100\ \mathrm{W\,m^{-2}}$; the atmosphere reflects 20 and the surface reflects 10. The atmosphere absorbs no solar radiation. It emits infrared intensities 25 downward and 15 upward, all in $\mathrm{W\,m^{-2}}$. What infrared intensity must it absorb from the surface?



A. $30\ \mathrm{W\,m^{-2}}$

B. $70\ \mathrm{W\,m^{-2}}$

C. $95\ \mathrm{W\,m^{-2}}$

D. $40\ \mathrm{W\,m^{-2}}$

1A Question 9 [1 mark]

A sealed sample of ideal gas expands to twice its original volume. Its pressure rises to 1.5 times its original pressure. What is the ratio of final to initial rms molecular speed?

A. 1.5

B. $\sqrt{3}$

C. $\sqrt{0.75}$

D. 3

1A Question 10 [1 mark]

An insulated mixture contains $20\,\text{g}$ of ice and $80\,\text{g}$ of water, both at 0°C. It receives $8.00\,\text{kJ}$. Use latent heat of fusion $3.30 \times 10^5\,\text{J kg}^{-1}$ and water specific heat capacity $4200\,\text{J kg}^{-1}\,\text{K}^{-1}$. What is the final temperature?

A. 0°C

B. 4.17°C

C. 19.0°C

D. 3.33°C

1A Question 11 [1 mark]

A cell of emf $18\,\text{V}$ and internal resistance $2.0\,\Omega$ supplies a $4.0\,\Omega$ resistor in parallel with a $12\,\Omega$ resistor. The $12\,\Omega$ branch is opened. What happens to the terminal potential difference and the current in the $4.0\,\Omega$ resistor?

A. Terminal potential difference decreases; the $4.0\,\Omega$ current increases.

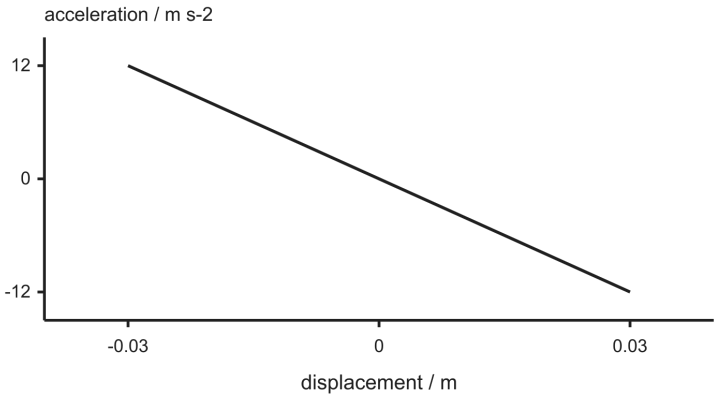
B. Both decrease.

C. Both increase.

D. Terminal potential difference increases; the $4.0\,\Omega$ current decreases.

1A Question 12 [1 mark]

The acceleration-displacement graph is for a mass m attached to an ideal spring. The spring is then used with mass $4m$. What is the new oscillation period?



A. $0.157\,\text{s}$

B. $1.26\,\text{s}$

C. $0.628\,\text{s}$

D. $0.314\,\text{s}$

1A Question 13 [1 mark]

Sound travels from air into water at an incidence angle of $10°$ to the normal. The speeds are $340\,\text{m}\,\text{s}^{-1}$ and $1480\,\text{m}\,\text{s}^{-1}$. What is the approximate refraction angle, and what happens to frequency?

A. $49°$; increased by a factor 4.35

B. $49°$; unchanged

C. $2.3°$; unchanged

D. $10°$; unchanged

1A Question 14 [1 mark]

A transparent solid has a critical angle of $30°$ at a solid-air boundary. The solid is immersed in liquid of refractive index 1.33. What is the new critical angle at the solid-liquid boundary?

A. $42°$

B. $62°$

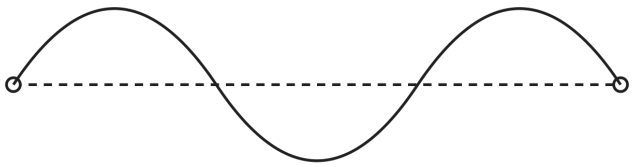
C. $22°$

D. $30°$

Paper 1A | Question paper

1A Question 15 [1 mark]

A uniform string fixed at both ends has fundamental frequency $50\ \text{Hz}$. Its vibrating length is increased by a factor 1.5 and its tension by a factor 2.25, without changing mass per unit length. What is its new third-harmonic frequency? Use the supplied relation $v = \sqrt{F/\mu}$, where F is tension and μ is mass per unit length.



third harmonic: three loops

A. $225\ \text{Hz}$

B. $337.5\ \text{Hz}$

C. $100\ \text{Hz}$

D. $150\ \text{Hz}$

1A Question 16 [1 mark]

A spectral line of rest wavelength $500.00\ \text{nm}$ is observed at $500.75\ \text{nm}$ in star X. A line of rest wavelength $650.00\ \text{nm}$ is observed at $650.65\ \text{nm}$ in star Y. Both speeds are much less than c. What is v_X/v_Y?

A. 0.667

B. 1.95

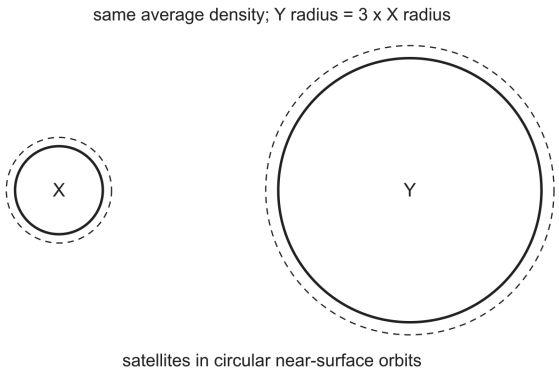
C. 1.50

D. 1.15

Paper 1A | Question paper

1A Question 17 [1 mark]

Two spherical planets have the same average density. Planet Y has three times the radius of planet X. A satellite moves in a circular orbit just above each planet's surface. Neglect satellite masses and the small orbital altitudes. What is T_Y/T_X?



A. $3\sqrt{3}$

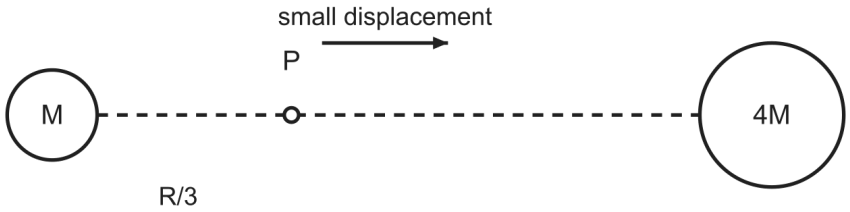
B. $\sqrt{3}$

C. 1

D. 3

1A Question 18 [1 mark]

Two isolated fixed masses M and $4M$ are separated by R. Their gravitational field is zero at P, between them and $R/3$ from M. A small test body at P is displaced slightly toward $4M$ and released. What is its initial acceleration and what does this imply about equilibrium along the line?



A. Toward $4M$; unstable

B. Zero; neutral

C. Toward $4M$; stable

D. Toward M; stable

1A Question 19 [1 mark]

A negative rod is brought near an initially neutral metal sphere without touching it. The sphere is grounded. In procedure X the ground is removed before the rod. In procedure Y the rod is removed while the ground remains, and the ground is removed last. What are the final sphere charges?

A. X positive; Y positive

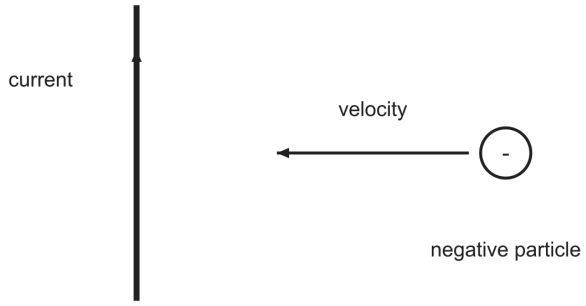
B. X positive; Y neutral

C. X negative; Y neutral

D. X neutral; Y negative

1A Question 20 [1 mark]

A long vertical wire carries current upward. A negatively charged particle lies to its right and moves horizontally toward the wire. What is the magnetic-force direction at that instant?



A. Into the page

B. Away from the wire

C. Upward, parallel to the current

D. Downward, opposite to the current

1A Question 21 [1 mark]

An atom has the three energy levels shown. A gas is illuminated by photons of energy $3.0\,\text{eV}$. Which change can make it absorb these photons, assuming the transitions between the shown levels are allowed?



three allowed levels

A. Lower the photon intensity while keeping all atoms in the ground state.

B. Populate the $2.0\,\text{eV}$ level before illumination.

C. Replace each photon by two separate $1.5\,\text{eV}$ photons and assume single-photon absorption.

D. Put every atom in the ground state.

1A Question 22 [1 mark]

In a binding-energy model, a model nucleus with 28 protons and 28 neutrons and binding energy per nucleon $8.8\,\text{MeV}$ is separated into fourteen helium-4 nuclei, each with binding energy per nucleon $7.1\,\text{MeV}$. Neglect kinetic energies. What net energy transfer is required?

A. $492.8\,\text{MeV}$ released

B. $95.2\,\text{MeV}$ supplied

C. $1.7\,\text{MeV}$ supplied

D. $95.2\,\text{MeV}$ released

1A Question 23 [1 mark]

A thermal-neutron fission reactor operates at constant power. Neutron-absorbing control rods are inserted farther while all other conditions initially remain unchanged. Which is the immediate cause of the reduction in power?

A. Fewer neutrons survive to cause further fissions.

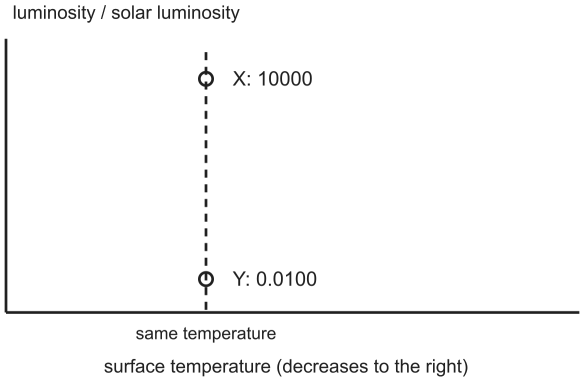
B. The rods slow neutrons mainly by elastic collisions without absorbing them.

C. The rods increase the mass of each uranium nucleus.

D. Each fission now releases less energy because the binding energies change.

1A Question 24 [1 mark]

Stars X and Y have the same surface temperature. Their luminosities are $10^4 L_\odot$ and $0.0100 L_\odot$, respectively. What is R_X / R_Y?



A. 1000

B. 100

C. 10^4

D. 10^6

1A Question 25 [1 mark]

Stars X and Y have equal luminosity. Their parallax angles are 0.050 and 0.200 arc-seconds. Interstellar absorption is negligible. What is the ratio of observed brightness b_X / b_Y?

A. $1/4$

B. 16

C. 4

D. $1/16$

1B Question 1 | Question paper

Paper 1B - Experimental and data-based questions

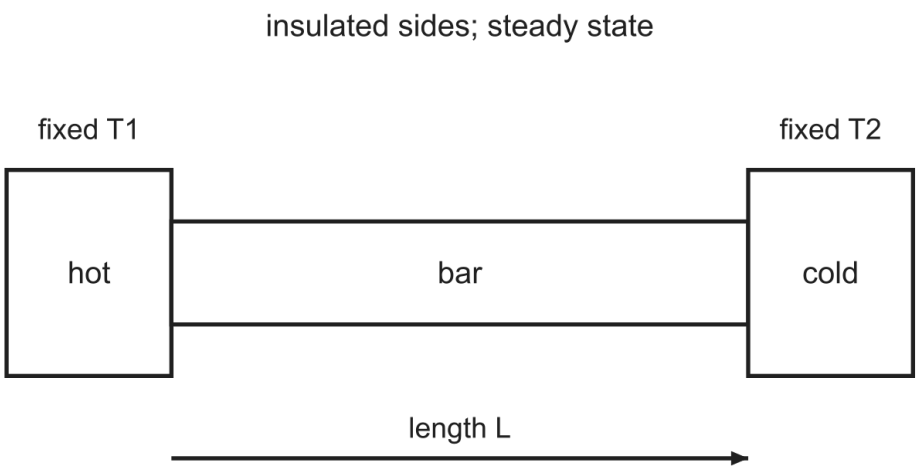
Continue within the same examination time. Answer both questions. Use the data, graphs and uncertainty information provided.

1B Question 1 [10 marks]

A laboratory measures the steady heater power P needed to maintain a temperature difference $\Delta T = (30 \pm 1)\,\mathrm{K}$ across bars of one material. Each bar has area $A = (2.00 \pm 0.04) \times 10^{-4}\,\mathrm{m}^2$. A constant parasitic power loss P_0 is proposed, so $P = kA\Delta T/L + P_0$.

Bar length L/m	0.100	0.150	0.200	0.300	0.400
Heater power P /W	9.00	7.00	6.00	5.00	4.50

A linear fit using a suitable horizontal-axis variable gives gradient $(0.600 \pm 0.030)\,\mathrm{W\,m}$.



(a) Choose a horizontal-axis quantity that linearizes the model. State the physical meanings of the gradient and vertical intercept. [2]

(b) Use the data to determine P_0, explaining why forcing the graph through the origin is inappropriate. [2]

(c) Calculate the thermal conductivity k. [2]

(d) Calculate the maximum percentage uncertainty in k, and state k with an appropriate absolute uncertainty. [2]

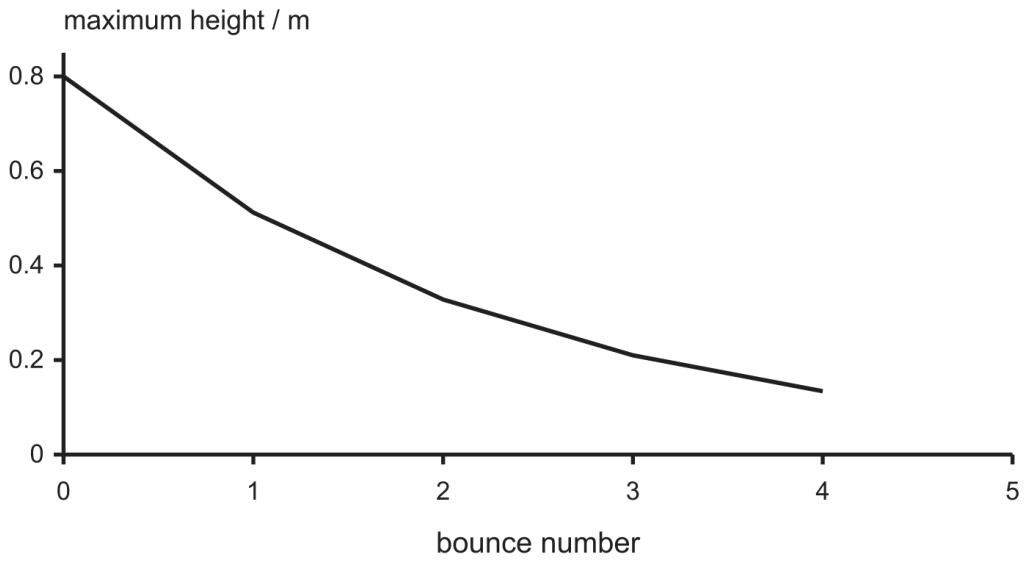
(e) Give one physical reason the constant-P_0 model may fail for bars of different length and one practical improvement. [2]

1B Question 2 | Question paper

1B Question 2 [10 marks]

A ball bounces vertically on a fixed floor. Air resistance between impacts is negligible. The recorded maximum heights are shown. The measurement uncertainty in every height is $0.005\ \text{m}$. A student proposes that the same fraction of mechanical energy is retained at each impact.

Bounce number	0: release	1	2	3	4
Maximum height / m	0.800	0.512	0.328	0.210	0.134



(a) Use the release and first rebound to find the fraction of mechanical energy lost at the first impact. [2]

(b) Find the ratio of the speed immediately after this impact to the speed immediately before it. [2]

(c) Using the constant-retention model, predict the fifth rebound height from the release height. [2]

(d) The measured fifth height is $(0.040 \pm 0.005)\ \text{m}$. A propagated model prediction is $(0.086 \pm 0.007)\ \text{m}$. Decide whether these data support the model, using the uncertainties. [2]

1B Question 2 | Question paper

(e) A common calibration error makes every measured height 10% too large. Explain its effect on the energy-retention fraction inferred from height ratios. [2]

End of question paper

PHYSICS

Standard Level

Set B | Paper 1A and 1B

Worked solution book

Complete the examination before consulting this book. Marking points indicate a method (M), accuracy (A) or reasoning (R) award. The number after the code gives its value. Award only points supported by the candidate's retained working.

Accept valid equivalent approaches and reasonable rounding. Apply error carried forward only where the subsequent method remains physically meaningful; do not reward a contradiction of stated conditions. A correct multiple-choice letter earns one mark; no negative marks apply.

MCQs are automatically marked on the portal. Written marks are self-assessed. Reports show these separately and do not calculate an official IB grade.

Paper 1A - Answer key and explanations

1	2	3	4	5	6	7	8	9	10
C	D	B	A	D	C	A	D	B	D

11	12	13	14	15	16	17	18	19	20
C	C	B	A	D	C	C	A	B	C

21	22	23	24	25
B	B	A	A	D

1A Question 1 [1 mark]

Correct answer: C

The distance in the first second is $4(1)+\frac{1}{2}(10)(1)^2=9\,\text{m}$. During $t=2$ to $3\,\text{s}$ it is $[4(3)+5(3)^2]-[4(2)+5(2)^2]=29\,\text{m}$. Equal velocity increments do not imply equal distances. Therefore choose C.

1A Question 2 [1 mark]

Correct answer: D

The vertical impact speed is $\sqrt{2gh}=\sqrt{360}=18.97\,\text{m s}^{-1}$. Horizontal speed remains $12\,\text{m s}^{-1}$. Thus $\tan\theta=18.97/12$, giving 57.7°. The displacement angle is not the velocity angle. Therefore choose D.

1A Question 3 [1 mark]

Correct answer: B

The signed acceleration-time area is $\frac{1}{2}(2)(6)+\frac{1}{2}(6-2)(4)=14\,\text{m s}^{-1}$. Therefore $F_{\text{av}}=m\Delta v/\Delta t=3(14)/6=7.0\,\text{N}$. The negative tail must be included with its sign. Therefore choose B.

1A Question 4 [1 mark]

Correct answer: A

Taking the lighter cart's motion positive, initial momentum is $mv-3mv=-2mv$. The combined velocity is $-v/2$. Initial energy is $\frac{1}{2}mv^2+\frac{1}{2}(3m)v^2=2mv^2$, and final is $\frac{1}{2}(4m)(v/2)^2=\frac{1}{2}mv^2$. The fraction is $1/4$. Therefore choose A.

1A Question 5 [1 mark]

Correct answer: D

Useful power is $\dot{m}(gh + v^2/2) = 12(50 + 2) = 624\,\text{W}$. Electrical input is $624/0.60 = 1040\,\text{W}$. Both potential and kinetic energy gains count as useful output. Therefore choose D.

1A Question 6 [1 mark]

Correct answer: C

For each rod, thermal resistance is $L/(kA)$. If the thin rod has resistance R, the thick rod has $R/4$. Initial series resistance is $5R/4$, while final resistance is $R/2$. At fixed total temperature difference, the rate increases by $(5R/4)/(R/2) = 2.5$. Therefore choose C.

1A Question 7 [1 mark]

Correct answer: A

Subtract $P = mc\dot{T} + H$ at the common temperature to cancel the same loss H: $40 = (0.20)c(8/60)$. Thus $c = 1500\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$. It is incorrect to assume all either heater's power warms the liquid. Therefore choose A.

1A Question 8 [1 mark]

Correct answer: D

Apply the energy balance to the atmosphere itself: its only absorption is surface infrared, and it emits $25 + 15 = 40\,\text{W}\,\text{m}^{-2}$. The surface emits $70 + 25 = 95$, of which 55 escapes directly and 40 is absorbed. The whole system then loses $55 + 15 = 70$, matching absorbed sunlight. Therefore choose D.

1A Question 9 [1 mark]

Correct answer: B

For fixed particle number, $T \propto PV$, so $T_2/T_1 = (1.5)(2) = 3$. Since $v_{\text{rms}} \propto \sqrt{T}$, the speed ratio is $\sqrt{3}$. The gas expansion alone does not imply cooling when energy is supplied. Therefore choose B.

1A Question 10 [1 mark]

Correct answer: D

Melting the ice requires $0.020(3.30 \times 10^5) = 6600\,\text{J}$. The remaining $1400\,\text{J}$ warms all $0.100\,\text{kg}$ of water. Hence $\Delta T = 1400/(0.100 \times 4200) = 3.33\,\text{K}$. The mass to warm includes the melted ice. Therefore choose D.

1A Question 11 [1 mark]

Correct answer: C

Initially external resistance is $3.0\,\Omega$, cell current $3.6\,\text{A}$, terminal voltage $10.8\,\text{V}$ and $4\,\Omega$ current $2.7\,\text{A}$. After opening, cell current is $18/6 = 3.0\,\text{A}$, terminal voltage $12\,\text{V}$ and branch current $3.0\,\text{A}$. Reduced internal voltage loss raises both requested quantities. Therefore choose C.

1A Question 12 [1 mark]

Correct answer: C

The slope gives $\omega^2 = 12/0.030 = 400\,\text{s}^{-2}$, so the original period is $2\pi/20 = 0.314\,\text{s}$. For the same spring, $T \propto \sqrt{m}$; quadrupling mass doubles the period to $0.628\,\text{s}$. Therefore choose C.

1A Question 13 [1 mark]

Correct answer: B

For waves $\sin\theta_1/\sin\theta_2 = v_1/v_2$, giving $\sin\theta_2 = (1480/340)\sin 10^\circ = 0.756$. Thus $\theta_2 \approx 49^\circ$. Boundary oscillations preserve frequency; the wavelength increases. Therefore choose B.

1A Question 14 [1 mark]

Correct answer: A

With air index 1, $n_s = 1/\sin 30^\circ = 2.00$. At the new boundary, $\sin c = n_l/n_s = 1.33/2 = 0.665$, so $c = 41.7^\circ$. The solid's refractive index has not changed; the index ratio has. Therefore choose A.

1A Question 15 [1 mark]

Correct answer: D

Wave speed is proportional to $\sqrt{F}$, so it increases by 1.5. Fundamental frequency is $v/(2L)$, and the same factor in length cancels this change. It remains $50\,\text{Hz}$; the third harmonic is $150\,\text{Hz}$. Therefore choose D.

1A Question 16 [1 mark]

Correct answer: C

For low speeds, $v/c = \Delta\lambda/\lambda_0$. The ratio is $(0.75/500)/(0.65/650) = 1.50$. Comparing shifts alone omits the different rest wavelengths. Both shifts are positive, so both sources recede. Therefore choose C.

1A Question 17 [1 mark]

Correct answer: C

Equal average density gives $M \propto R^3$. For a near-surface circular orbit, $T^2 = 4\pi^2 R^3/(GM)$. The ratio R^3/M is the same for both planets, so their periods are equal. Larger radius alone does not imply a longer period when the central mass also changes. Therefore choose C.

1A Question 18 [1 mark]

Correct answer: A

Moving toward $4M$ strengthens that mass's attraction and weakens the attraction toward M. The resultant therefore points in the displacement direction, toward $4M$, so the body moves farther from P. The equilibrium is unstable along this line. Therefore choose A.

1A Question 19 [1 mark]

Correct answer: B

The negative rod repels electrons from the sphere into Earth. Removing the ground first traps an electron deficit, so X is positive after the rod is removed. If the rod is removed while still grounded, electrons return until the sphere is neutral; Y then remains neutral. Order is essential. Therefore choose B.

1A Question 20 [1 mark]

Correct answer: C

The right-hand grip rule gives a magnetic field into the page to the right of the wire. For positive charge, leftward velocity crossed with an into-page field gives a downward force. The negative charge reverses this to upward, parallel to the wire current. Therefore choose C.

1A Question 21 [1 mark]

Correct answer: B

A $3.0\,\text{eV}$ photon matches the $5.0 - 2.0\,\text{eV}$ gap. The lower state of that transition must be populated. From the ground state the available gaps are 2.0 and $5.0\,\text{eV}$, so $3.0\,\text{eV}$ is not absorbed in this three-level model. Intensity alone does not change photon energy. Therefore choose B.

1A Question 22 [1 mark]

Correct answer: B

Initial total binding is $56(8.8) = 492.8\,\text{MeV}$. Final binding is $14(4)(7.1) = 397.6\,\text{MeV}$. Decreasing binding by $95.2\,\text{MeV}$ requires that energy to be supplied; greater binding corresponds to lower rest energy. Therefore choose B.

1A Question 23 [1 mark]

Correct answer: A

Control rods absorb neutrons, lowering the number available to continue the chain reaction and reducing the fission rate. Moderation mainly slows neutrons; it is a different role. The nuclear energy released per fission is not changed by rod position. Therefore choose A.

1A Question 24 [1 mark]

Correct answer: A

Stefan-Boltzmann gives $L = 4\pi R^2 \sigma T^4$. With the same T, the radius ratio is the square root of the luminosity ratio: $\sqrt{10^4/0.0100} = \sqrt{10^6} = 1000$. Vertical separation on an HR diagram at fixed temperature can correspond to very different radii. Therefore choose A.

1A Question 25 [1 mark]

Correct answer: D

Distance is inversely proportional to parallax, so $d_X/d_Y = 0.200/0.050 = 4$. Apparent brightness for equal luminosity follows $b \propto 1/d^2$, giving $b_X/b_Y = 1/16$. Larger parallax means nearer and brighter in this comparison. Therefore choose D.

Paper 1B - Worked solutions

1B Question 1 [10 marks]

(a) [2 marks]

M1 Plot P against $1/L$, measured in m^{-1}.

A1 The gradient is $kA\Delta T$ and the vertical intercept is parasitic power P_0.

(b) [2 marks]

M1 For example, use $P_0 = 9.00 - 0.600(1/0.100)$.

A1 $P_0 = 3.00\,\mathrm{W}$. A nonzero loss exists even in the linearized model, so forcing zero intercept misassigns loss to conduction.

(c) [2 marks]

M1 $k = 0.600/(2.00 \times 10^{-4} \times 30)$.

A1 $k = 100\,\mathrm{W\,m^{-1}\,K^{-1}}$.

(d) [2 marks]

M1 $\delta k/k = 0.030/0.600 + 0.04/2.00 + 1/30 = 0.1033$.

A1 The maximum percentage uncertainty is about 10%, giving $k = (100 \pm 10)\,\mathrm{W\,m^{-1}\,K^{-1}}$.

(e) [2 marks]

R1 For example, longer bars may have more exposed surface and therefore different lateral heat loss; contact resistance may also differ between assemblies.

R1 Use effective insulation/guard heating to reduce lateral loss, or a controlled contact method to reduce variable contact resistance. The improvement must address the stated limitation.

1B Question 2 [10 marks]

(a) [2 marks]

M1 Energy at a turning point is mgh. The retained fraction is $0.512/0.800 = 0.640$.

A1 The fraction lost is $1 - 0.640 = 0.360$, or 36.0%.

(b) [2 marks]

M1 Kinetic energy is proportional to speed squared; use $v_{\text{after}}/v_{\text{before}} = \sqrt{0.640}$.

A1 The speed ratio is 0.800.

(c) [2 marks]

M1 The fifth rebound follows five impacts, so $h_5 = 0.800(0.640)^5$.

A1 $h_5 = 0.0859\,\text{m}$, approximately $0.086\,\text{m}$.

(d) [2 marks]

R1 The measured interval $[0.035, 0.045]\,\text{m}$ and predicted interval $[0.079, 0.093]\,\text{m}$ do not overlap.

R1 The discrepancy is larger than the stated combined bounds, so the later result does not support constant retention over all five impacts. It does not identify a unique cause of failure.

(e) [2 marks]

M1 Each ratio becomes $(1.10h_{n+1})/(1.10h_n)$.

R1 The common factor cancels, so the inferred retention fraction is unchanged. Absolute heights and energy estimates would still be biased.

End of worked solution book